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Game Theoretical Bandwidth Allocation in UAV-UGV Collaborative Disaster Relief Networks

Bincheng Ying^{+}, Zhou Su^{++}, Qichao Xu^{+}, Xiandong Ma^{+}
+School of Mechatronic Engineering and Automation, Shanghai University, China
++School of Cyber Science and Engineering, Xi'an Jiaotong University, China
Corresponding Author: zhousu@ieee.org

***Abstract*—After a natural disaster, cellular communications may be hard to meet the needs of disaster relief due to the destruction of infrastructures, such as base stations and power stations. In order to transmit rescue data in time, the disaster relief network based on unmanned ground vehicles (UGVs) is advocated to transmit rescue-critical data to the relief center. However, due to the driving-difficult roads in the disaster area and small communication coverage of UGVs, UGVs-assisted disaster relief network faces the challenge to quickly and accurately collect wide-range disaster information. To tackle this problem, in this paper, we first take unmanned aerial vehicles (UAVs) to develop a UAV-UGV collaborative disaster relief network architecture, where the UAVs with high mobility, wide view and agility can efficiently help UGVs to perceive the disaster information. Then, to realize the efficient data delivery between UAVs and UGVs with limited bandwidth resources, we devise a game-based bandwidth allocation framework. In specific, the multi-leaders multi-followers Stackelberg game is used to model the interactions between UGVs and UAVs, where each UGV acts as a leader to determines its bandwidth capacity and each UAV acts as a follower to determines its access selection based on the bandwidth capacity. Especially, the non-cooperative game model is introduced to analyze the competitions among UGVs, and the evolutionary game is utilized to study the strategy adjustments of UAVs. Through the backward induction method, the optimal access selection of each UAV and the optimal bandwidth capacity of each UGV are obtained, which form the Stackelberg equilibrium. At last, extensive simulation results demonstrate that the proposed scheme outperforms the conventional counterparts on the utilities of both UAVs and UGVs.**

***Index Terms*—Disaster relief network, UAV-UGV collaboration, bandwidth allocation, Stackelberg game, evolutionary game.**

I. Introduction

As it is known to all, natural disasters often cause a large number of innocent human casualties and economic losses in the world [1], [2]. Therefore, how to minimize the loss of economy and lives after disaster has become a great challenge for countries. The restoration of communication networks plays a critical role in disaster relief [3]. If the communication networks are failed, the working competence of rescue teams would degrade rapidly. According to the previous work [4], the communication network should be rebuilt in 72 hours as soon as possible to establish effective and efficient connections between rescue workers and the remote command center.

However, because of the unpredictable natural disasters, the original communication infrastructures may be damaged seriously, which cannot support the efficient post-disaster rescue. As such, how to rapidly deploy emergency communication networks in such disaster areas has been of a great concern problem. Nowadays, some schemes combined with device-to-device (D2D) communication and wireless sensors networks (WSNs) [5] have been proposed to build disaster relief networks in disaster areas. However, the communication distance between two wireless nodes is short and the communication performance can be easily affected by geography. In addition, wireless nodes in the affected areas are easily damaged by secondary disasters.

In recent years, unmanned ground vehicles (UGVs) equipped with a variety of sensors, large-scale batteries, and communication modules can replace human beings to enter disaster areas for carrying out dangerous missions with long endurance. Therefore, UGVs equipped with communication modules can be employed to establish disaster relief networks. However, the roads in the disaster regions can be damaged seriously which leads to that in UGVs can not enter deep into the affected zones. Besides, due to the limitation of mobility and communication coverage, UGVs are not able to serve a large range of devices effectively.

Due to the high flexibility, wide communication coverage, and light-of-sight (LoS), unmanned aerial vehicles (UAVs) can transmit data to ground devices [6]. Through air-to-ground (A2G) and air-to-air (A2A) communications, UAVs can be adopted to build air-ground networks with high quality. Therefore, the application of UAVs is an effective solution to assist UGVs-based networks in disaster areas. UAVs can utilize the bandwidth shared by UGVs to serve the ground communication equipment. However, there are still several challenges for UAVs and UGVs to build an efficient emergency communication network. First, each UGV has no idea about the capacity of bandwidth it should share with UAVs. Second, UAVs usually select a UGV sharing a large bandwidth capacity in order to maximize the transmission rate, which will result in a UGV selected by too many UAVs at the same time. Furthermore, the bandwidth is limited under disaster situations. Therefore, it is necessary to design an efficient scheme to allocate bandwidth in the disaster relief network.

In this paper, we propose a game theoretical bandwidth allocation in UAV-UGV collaborative disaster relief network to transmit rescuing data. Firstly, we jointly employ the UAVs and UGVs to develop a UAV-UGV collaborative disaster relief network architecture. Secondly, in order to solve the problem of delivering data with limited bandwidth resources, we devise a game-based bandwidth allocation framework. Thirdly, an iterative algorithm based on backward induction is developed to obtain the Stackelberg equilibrium. Finally, extensive simulation results prove that our scheme has better performance than conventional counterparts on the utilities of both UAVs and UGVs. In a nutshell, the main contributions of this work are as follows:

- **Disaster Relief Network Architecture:** We novelly develop a UAV-UGV cooperative disaster relief network architecture based on UAVs and UGVs equipped with communication modules. The UGVs move along the roads damaged slightly and deicide the shared bandwidth capacity. Then, the UAVs access the UGVs to use the shared bandwidth to serve the disaster communication devices.
- **Bandwidth Allocation Scheme:** In order to transmit rescuing data between UGVs and UAVs with limited bandwidth, we investigate a game-based bandwidth allocation framework. The Stackelberg game is used to model the interactions between UGVs and UAVs, where UGVs are leaders who decide the shared bandwidth capacity and UAVs are followers who adjust access selection strategy. Besides, we utilize the non-cooperative game to represent the competition among UGVs and the evolutionary game to indicate the access selection of UAVs. Furthermore, through the backward induction method, an iterative algorithm is proposed to attain the Stackelberg equilibrium.
- **Simulation Evaluation:** We evaluate the effectiveness and efficiency of our proposed scheme with plenty of simulations. The results prove that our scheme can motivate participants to select access points legitimately, effectively allocate bandwidth resources, and bring higher utilities for both UAVs and UGVs than the conventional counterparts.

The rest of this paper is organized as follows. We describe related work in Section II. Section III presents the system model. In Section IV, the problem formulation is described in detail. Section V shows the game analysis of UGVs and UAVs. Simulation results are shown in Section VI and we conclude the paper in Section VII.

II. Related Work

Recently, UAVs are widely adopted for post-disaster rescue in the affected areas. Yang *et al*. [7] formulated data transmission problem by utilizing multi-UAVs in disaster areas and optimized the trajectories of UAVs, task completion time, and energy consumption. An adaptive arrangement scheme of UAVs was proposed by Nan *et al*. to resolve the coverage problem in UAV-aided GNs communication networks to achieve the maximum coverage [8]. In [9], Wang *et al*. proposed that the emergency information can be transferred from the UAVs to ground users in a multicast mechanism in the disaster relief network.

There are several resource allocation schemes in communication networks. Li *et al*. first express the significance of allocating bandwidth appropriately for stream-reservation traffic under the Credit-Based Shaping and construct a general

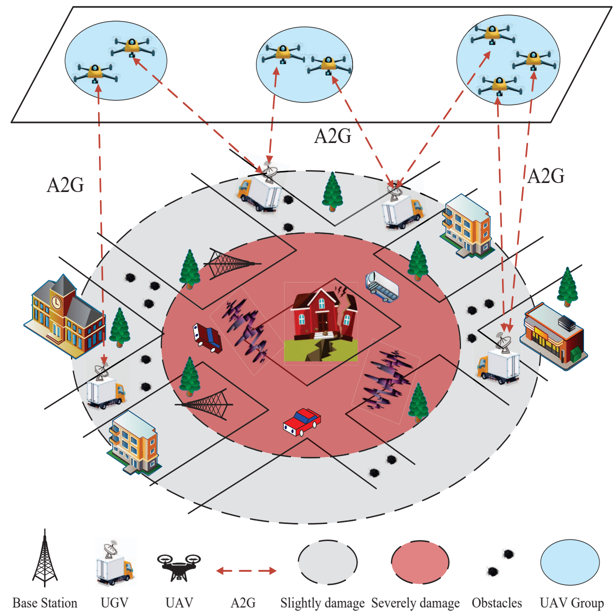

Fig. 1. UAVs and UGVs cooperated emergency communication network in the disaster area.

schema for its bandwidth allocation in [10]. A stochastic optimization problem of jointly selecting client and allocating bandwidth with energy constraints of long-term client was proposed in [11]. Besides Xu *et al*. [11] developed an algorithm that employs available wireless channel information currently but guarantee long-term performance. In [12], a bandwidth allocation methodology was proposed with considering the inter-beam interference by Kawamoto *et al*. to allocate the resources of the satellite communication system.

However, most of bandwidth allocation schemes can not be applied in the disaster relief network directly. Some of above networks are established only by UAVs. Due to the limitation of battery capacity and communication power, it's not practical that UAVs working for a long time and transmit data to remote control center. Others heavily rely on terrestrial base stations, which can be damaged seriously due to the natural disaster. In our work, we build the disaster relief network with UAVs and UGVs. UGVs equipped with communication modules are introduced to take over the work of base stations. The advantages of these two kinds of unmanned vehicles complement each other.

III. System Model

In this section, we present the system model including UGVs motion model, UAVs motion model and communication model. UGVs equipped with communication modules can be deployed in the disaster areas. UAVs can provide communication services to the ground devices.

A. UGVs Motion Model

As shown in Fig. 1, we consider the disaster relief network consisting I UGVs and J UAVs. Let $\mathbb{I} = \{1, 2, ...i, ..., I\}$ to denote the set of UGVs. UGVs can drive automatically along the predefined and damaged slightly path in the disaster area. Here, we adopt a time slot model [13] [14], each time slot is equal and the length is τ. t_k represents k-th slot. At time slot t_k, the velocity of i-th UGV is $\boldsymbol{v}_{i,k}$. And the maximum and minimum velocity of UGV i are $v_i^{\max}$ and $v_i^{\min}$, respectively.

Since there exist some puddles and cracks on the roads, UGVs should avoid these obstacles. The length of road with η lanes that UGV i travels along is L_i. The area of road L_i is S_i. Similar to [15], the spatial distribution of obstacles follows a Poisson Point Process (PPP) distribution with a constant density λ. Hence, the number of obstacles on each lane is $\frac{\lambda S_i}{\eta}$. In the moving process, UGVs would follow a circular path to avoid obstacles with radius $R_i (R_i > 0)$. In order to simplify the calculations, the impact of obstacles avoidance on the time of UGVs is transformed into the impact on UGVs' velocities accordingly:

$$\boldsymbol{v}'_{i,k} = \frac{\boldsymbol{v}_{i,k} \eta L}{\left[L + (\pi R_i - 2R_i)\right] \lambda S_i}. \tag{1}$$

The trajectory of i-th UGV can be expressed as $\boldsymbol{q}_i = [q_{i,0}, q_{i,1}, q_{i,2}, ..., q_{i,k}, ...]$, where the beginning position is $q_{i,0} = (X_{i,0}, Y_{i,0}, 0)$. And $q_{i,k} = (X_{i,k}, Y_{i,k}, 0)$ denotes the position of i-th UGV at time slot t_k. The movement of UGV i satisfies the following constraints:

$$\| q_{i,k} - q_{i,k-1} \| = \tau \boldsymbol{v}'_{i,k}, \forall i \in \mathbb{I}, \tag{2}$$

$$v_i^{\min} \leqslant \| \boldsymbol{v}_{i,k} \| \leqslant v_i^{\max}, \forall i \in \mathbb{I}. \tag{3}$$

B. UAVs Motion Model

UAVs can connect to the ground UGVs through Orthogonal Frequency Division Multiplexing Access (OFDMA) technology. Each UAV connects to at most one UGV in each time slot, but one UGV can be accessed by multiple UAVs simultaneously. The set of UAVs is denoted as $\mathbb{J} = \{1, 2, ..., j, ..., J\}$. We can divide all UAVs into many groups according to different mission features [17]. Let $\mathbb{G} = \{1, 2, ..., g, ..., G\}$ denote the set of UAV groups, where the number of UAV groups is G. There are N_g UAVs in g-th UAV group. The working altitude of UAV j in group g is $H_{j,k}^g$, the cruising velocity is $\boldsymbol{v}_{j,k}^g$, the maximum and minimum velocity are $v_j^{\max}$ and $v_j^{\min}$, respectively. We denote $q_{j,0}^g = \left(X_{j,0}^g, Y_{j,0}^g, H_{j,0}^g\right)$ as beginning position of j-th UAV in group g and $q_{j,k}^g = \left(X_{j,k}^g, Y_{j,k}^g, H_{j,k}^g\right)$ as the position of j-th UAV in group g at time slot t_k. The trajectory of j-th UAV is expressed by $\boldsymbol{q}_j = \left[q_{j,0}^g, q_{j,1}^g, q_{j,2}^g, ..., q_{j,k}^g, ...\right]$. Therefore, the motion constraints of j-th UAV can be expressed by

$$\| q_{j,k}^g - q_{j,k-1}^g \| = \tau \boldsymbol{v}_{j,k}^g, \forall j \in \mathbb{J}, \tag{4}$$

$$v_j^{\min} \leqslant \| \boldsymbol{v}_{j,k}^g \| \leqslant v_j^{\max}, \forall j \in \mathbb{J}. \tag{5}$$

C. Communication Model

UAVs and UGVs connect with each other through the air-to-ground (A2G) link. Since UAVs utilize the bandwidth of UGVs to provide communication services, each UAV needs to connect to a UGV. The distance between UGV i and UAV j can calculated by

$$d_{i,j,k}^g = \sqrt{\| q_{i,k} - q_{j,k}^g \|^2}. \tag{6}$$

At the k-th time slot, the connection status between UAVs and UGVs can be denoted as $a_{i,j,k}$, which is a binary parameter. If a connection is established between the i-th UGV and j-th UAV, $a_{i,j,k} = 1$, otherwise $a_{i,j,k} = 0$. Therefore, we can obtain $\sum_{i=1}^{I} a_{i,j,k} \leqslant 1$.

The transmission power of each UAV is P_j. There is no co-channel interference among UAVs in the different groups accessing the same UGV. Therefore, the signal-to-noise ratio (SNR) is

$$h_{i,j,k} = \frac{P_j \beta_{ij,k}^g}{\sigma^2}, \tag{7}$$

where $\beta_{i,j,k}^g = \frac{\beta_0}{\left(d_{i,j,k}^g\right)^2}$ is channel gain, σ^2 means the additive white Gaussian noise at the receiver, and β_0 represents the channel gain when the reference distance is 1m.

Supposing that each UGV is equipped with a single antenna, and the bandwidth capacity that UGV i decides to share is B_i. In order to ensure fairness, each UGV divides the bandwidth equally among the connected UAVs. Therefore, the transmission rate between i-th UGV and j-th UAV is

$$r_{i,j,k}^g = a_{i,j,k} \frac{B_i}{N_{i,k}} \log_2 \left(1 + h_{i,j,k}\right), \tag{8}$$

where $N_{i,k}$ represents the number of UAVs connecting to UGV i at time slot t_k. Additionally, we have $\frac{B_i}{N_{i,k}} \geqslant b_{\min} > 0$, where $b_{\min}$ is the bandwidth threshold.

IV. Problem Formulation

In this section, we propose a bandwidth allocation scheme among UAVs and UGVs. Firstly, a bandwidth allocation framework is introduced. Secondly, we formulate the utility functions of each player. Then, we show the details of the resource allocation based on the Stackelberg game.

A. Bandwidth Allocation Framework

In the disaster area, UGV i decides the capacity of bandwidth B_i for sharing to UAVs in the emergency communication network, and then announces its decision to UAVs. Meanwhile, according to the shared bandwidth capacity, UAVs select their own access points, i.e. UGVs.

To enable UAVs and UGVs cooperate with each other to maximize the transmission data rate for the entire communication network, a bandwidth allocation framework is presented. Specifically, the interaction between UGVs and UAVs can be formulated by a two-layer Stackelberg game. Firstly, UGVs determine the capacity of bandwidth resources shared and announce them to UAVs. Note that, the capacity of resource which has a great impact on both network performance and overall cost should be considered carefully. If the shared bandwidth capacity of a UGV is too large, a great number of UAVs will select it as the access point and the network performance will be degraded. Then, after obtaining the information of shared bandwidth capacity , each UAV can select an appropriate UGV as the access point for the sake of maximal network performance.

B. Utility Functions

The Utility Functions of UAVs: The objective of UAVs is to maximize the transmission rate of the disaster relief network due to the emergency of rescue-critical data. Therefore, they do not take costs into account that they should pay. The utility function of UAV j in group g can be defined as

$$u_{j,k}^{g} = \epsilon \log\left(1 + r_{ij,k}^{g}\right), \tag{9}$$

where ϵ is a parameter of the payoff function. According to Eq. (8), the utility of a UAV can be indicated as

$$u_{j,k}^{g} = \epsilon \log\left(1 + a_{ij,k} \frac{B_i}{N_{i,k}} \log_2\left(1 + h_{ij,k}\right)\right). \tag{10}$$

The Utility Functions of UGVs: In general, UGVs in the disaster area act as operators. Therefore, UGVs should consider the revenue. Each UGV predefines the price p_i that each established UAV must pay. In this way, each UGV can obtain revenue from the connected UAVs. In addition to this, the operating cost of a UGV also should be taken into account. Let C_i denote the operating cost of UGV i. Therefore, the utility function of UGV i can be defined as

$$U_i\left(B_i\right) = p_i N_i\left(B_i\right) - C_i, \tag{11}$$

where $N_i\left(B_i\right)$ denotes the number of UAVs connecting to the UGV i according to bandwidth capacity B_i. When UAVs do not change their strategies of access points, $N_i\left(B_i\right) = N_i$.

C. Stackelberg Game Based Resource Allocation

UAVs and UGVs want to obtain the optimal strategies, which means that a UGV adaptively changes its resource allocation decision to stimulate more UAVs for maximizing its utility. And a UAV appropriately selects a UGV as the access point according to the capacity of shared bandwidth to maximize the network transmission rate. Therefore, the bandwidth resource allocation problem can be formulated as a two-layer Stackelberg game by taking the utilities of UAVs and UGVs into account.

In the top layer, UGVs are the leaders in the Stackelberg game and share a real-time bandwidth resource B_i to UAVs. In the bottom layer, UAVs are the followers and rationally select UGVs on the ground as access points based on the capacity of shared bandwidth resource.

V. Game Analysis

In this section, we solve the optimization problems of UAVs and UGVs by the proposed Stackelberg game. The strategies in both layers have an effect on another layer. Similar to [18], the backward induction method is adapted in this section to analyze the Stackelberg game. Besides, access selection process of UAVs is described by the evolutionary game. After obtaining the evolutionary game result of UAVs, UGVs in the disaster area compete with each other and make appropriate bandwidth sharing strategies by a non-cooperative game.

A. Evolutionary Game Analysis of UAVs

Each UAV selects the access point according to the transmission rate and bandwidth capacity until all the UAVs in the same group have equal utility. Each UAV can only get the information from the neighbouring UAVs because of the limitation of computing capability and hardware resources. Therefore, the evolutionary game is regarded as an effective methodology to describe the process of UAVs' access points selection.

The access selection process of UAVs can be presented as follows:

1) Players : The players in the evolutionary game are all the UAVs in the disaster area.
2) Strategies : The strategies of UAVs are the selection of UGVs on the ground, which can be denoted as $S = \{i|i \in \mathbb{I}\}$.
3) Groups : Let the set $\mathbb{G}$ denote the groups of UAVs in disaster area $\mathbb{G} = \{1, 2, ..., g, ..., G\}$, and the number of UAVs in the g-th group is N_g.
4) Group state : For the g-th UAV group, the group state can be denoted as
$$x_i^g = N_i^g / N_g, \tag{12}$$
which means the proportion of UAVs selecting the i-th UGV as the access point in g-th group. N_i^g indicates the number of UAVs in group g selecting i-th UGV and
$$\sum_{i=1}^{I} x_i^g = 1. \tag{13}$$
5) Utility function : The utility function is used to express the Quality of Service (QoS) of j-th UAV who selects i-th UGV. Here, as mentioned before in this paper, we use Eq. (8) to express the utility function.
$$u_{j,k}^{g} = \epsilon \log\left(1 + a_{ij,k} \frac{B_i}{N_{i,k}} \log_2\left(1 + h_{ij,k}\right)\right). \tag{14}$$

From the above analysis, a UAV can obtain a better utility if it chooses a UGV sharing larger bandwidth, resulting in high transmission rate. However, if a UGV decides to share much larger bandwidth to UAVs, there will be much more UAVs select the same UGV, which will degrade the transmission rate rapidly.

Similar to the existing study [19], we adopt the replicator dynamic to solve the problem of access points selection by the replicator dynamic which is shown as following

$$\dot{x}_{i,k}^{g} = \xi x_{i,k}^{g}\left(u_{j,k}^{g} - \bar{u}_{i,k}^{g}\right), \tag{15}$$

where ξ is a positive parameter that is used to control the speed of UAVs' access selection in the groups. In addition, the average utilities of UAVs in g-th group can be calculated by

$$\bar{u}_{i,k}^{g} = \sum_{i=1}^{I} x_{i,k}^{g} u_{j,k}^{g}. \tag{16}$$

From the Eq. (15), if the utility of a UAV selecting UGV $i\,(i \in \mathbb{I})$ is greater than the average utility of the entire group, the replicator dynamic will be greater than zero. Then, the number of UAVs connecting with UGV $i\,(i \in \mathbb{I})$ will increase rapidly. Otherwise, if a certain UAV's utility is smaller than the average utility of the entire group, there will be fewer UAVs selecting this UGV as their access points. Therefore, at the end of the dynamic process of the evolutionary game, each UAV in the same group will have equal utility. And the utility of each UAV is equal to the average utility of the entire group, that is $u_{j,k}^{g} = \bar{u}_{i,k}^{g}$.

At this point, the x_i^g* corresponding to this group's state is Evolutionary Equilibrium (EE), which means that the players do not change their strategies by their own volition.

In order to estimate the stability of x_i^g*, the Jacobian matrix for the replicator dynamic should be estimated. The EE is Evolutionary Stable Strategy (ESS) if the eigenvalues of the Jacobian matrix are negative. Then, the solution of UAVs' access points selection can be obtained by

$$\mathbf{x}_i^g* = \left(x_1^g*, x_2^g*, .., x_i^g*, ..., x_I^g*\right). \tag{17}$$

B. Non-cooperative Game of UGVs

After obtaining the evolutionary game result of UAVs' access selection process, UGVs in the disaster area compete with each other. A non-cooperative game is adapted to describe the UGVs' competition in the disaster relief network. Then, the unique Nash equilibrium obtained from the game is regarded as the optimal solution.

Without loss of generality, a UGV determines whether it needs to change its own strategy until the number of UAVs connecting to itself no longer changes. Since UGVs cannot aware of the duration of UAVs evolution, we set the waiting time $T = a\tau$ for UGVs' strategy update progress, where a is a parameter to control the waiting time. According to Eq. (11), we can obtain the utility function of UGV i in non-cooperative game as following

$$U_i\left(\mathbb{B}_i, \mathbb{B}_{-i}\right) = U_i\left(B_i\right) = p_i N_i\left(B_i\right) - C_i, \tag{18}$$

where $\mathbb{B}_{-i} = \left(B_1, B_2, ..., B_{i-1}, B_{i+1}, ..., B_I\right)$ denotes other UGVs' bandwidth allocation strategies except i-th UGV. Due to the evolutionary game result of UAVs, $N_i\left(B_i\right)$ can be denoted as

$$N_i\left(B_i\right) = \sum_{g \in \mathbb{G}} N_g x_i^g*, \tag{19}$$

which means the total number of UAVs selecting the i-th UGV when the evolutionary game achieves equilibrium in the entire network. Therefore, the specific utility function of the i-th UGV can be indicated as

$$U_i\left(\mathbb{B}_i, \mathbb{B}_{-i}\right) = p_i \sum_{g \in \mathbb{G}} N_g x_i^g* - C_i. \tag{20}$$

In this paper, the Nash equilibrium of this non-cooperative game of UGVs' competition can be obtained by the best response function. Therefore, when given other UGVs' optimal strategies, the optimal strategy of i-th UGV can be expressed by

$$\mathcal{F}\left(\mathbb{B}_{-i}\right) = arg \max_{\mathbb{B}_i} U_i\left(\mathbb{B}_i, \mathbb{B}_{-i}\right). \tag{21}$$

Let $\mathbb{B}^* = \left(B_1*, B_2*, ..., B_I*\right)$ denote the Nash equilibrium of UGVs' non-cooperative game. The Nash equilibrium of this non-cooperative game can be obtained by solving the following equation

$$\mathcal{F}\left(\mathbb{B}_{-i}*\right) = B_i*. \tag{22}$$

In order to perform the above calculation, each UGV needs to obtain the evolutionary game equilibrium solution of all UAV groups. However, this is not implemented for UGVs in the realistic disaster area. Therefore, an efficient iterative method is adopted to adjust the bandwidth resource allocation for UGVs in the direction of increasing UGVs' utilities. Here, i-th UGV updates its capacity of bandwidth resource by the following equation

$$B_i\left(T+1\right) = B_i\left(T\right) + w_i \frac{dU_i\left(\mathbb{B}_i\left(T\right)\right)}{d\mathbb{B}_i\left(T\right)}, \tag{23}$$

where $B_i\left(T\right)$ represents the bandwidth capacity that i-th UGV decides to share during time T. Moreover, w_i is an update parameter which is adopted to control the speed of bandwidth sharing. For each UGV, the bandwidth resource allocation strategy is updated effectively by marginal payoff $\frac{d(U_i(B_i))}{d(B_i)}$, it can be calculated by its own small variable $\upsilon\left(eg: \upsilon = 10^{-4}\right)$:

$$\begin{aligned} &\frac{dU_i\left(\mathbb{B}_i\left(T\right)\right)}{d\mathbb{B}_i\left(T\right)} \approx \\ &\frac{U_i\left(..., B_i\left(T\right)+\upsilon, ...\right) - U_i\left(..., B_i\left(T\right)-\upsilon, ...\right)}{2\upsilon}. \end{aligned} \tag{24}$$

An iterative algorithm is proposed to obtain the optimal solution for our bandwidth allocation scheme shown in Algorithm 1. Specifically, the process of bandwidth allocation consists of two stages: 1) Access points selection. 2) Bandwidth allocation. In the stage 1, the UAVs adaptively select the UGVs as their access points by the evolutionary game in lines 3-20. In the stage 2, the UGVs allocate their bandwidth capacity according to the number of UAVs connected to them in lines 21-23.

TABLE I
SIMULATION PARAMETERS

Parameters	Values	Parameters	Values
G	2	I	3
η	1	R	1m
H_j	500m	σ^2	-174dBm/Hz
P_j	20W	B	40MHz
υ	1	ξ	1
$\{p_1, p_2, p_3\}$	$\{3, 2, 1\}$	$\{C_1, C_2, C_3\}$	$\{15, 10, 5\}$
T	100		

VI. Simulation Evaluations

In this section, the performance evaluation of our proposed scheme is presented in detail. First, we introduce the scenario parameters that we used in simulation. Then, we show the simulation results and analyses.

A. Simulation Setup

In the simulation, the disaster scenario consists of $G = 2$ UAVs groups and $I = 3$ UGVs. We set the road with a

Algorithm 1 : Bandwidth Resource Allocation Algorithm

Input: $\boldsymbol{v}_i$; $\boldsymbol{v}_j$; $i \in \mathbb{I}$; $j \in \mathbb{J}$; $g \in \mathbb{G}$; B; ϵ; ξ; L_i; η; S_i; λ; P_j; p_i
Repeat
while $t < T$ **do**
  Each UAV randomly selects a UGV as an access point to serve the communication device;
  for $g = 1 : 1 : G$ **do**
    while $u_{ij}^g(t_k) \neq \bar{u}_i^g(t_k)$ **do**
      for $J = 1 : 1 : J$ **do**
        Calculate the transmission rate for each UAV according to Eq. (8);
        Calculate the utility of each UAV in g-th group according to Eq. (9);
      end for
      Calculate the average utility of g-th UAV's group according to Eq. (16);
      The UAV in g-th group gets information of other UAVs in the same group;
      if $u_{ij}^g(t_k) < \bar{u}_i^g(t_k)$ **then**
        $i = i'$, as long as $u_{ij}^g(t_k) < u_{i'j}^g(t_k)$;
      else
        UAV i does not change its access point;
      end if
    end while
  end for
end while
Update the bandwidth capacity of UGVs sharing according to Eq. (23) and Eq. (24);
$t = t + 1$
Until B_i does not change;
Output: x_i^g; $U_i(B_i)$; u_j^g;

single lane. The avodance radius of obstacle is set as $R = 1$m. For the UAVs, we suppose their operating altitudes are equal and are set to be 500m. The additive white Gaussian noise in the communication links is -174dBm/Hz. Besides, the transmission power of each UAV is set to be equal and $P_j = 20$W. The total bandwidth resource is set to be 40MHz [19]. For convenience, the main parameters in the simulation are listed in Table I. Besides, we show the convergence of the proposed scheme. Moreover, we compare our scheme with two schemes: the scheme is that UAVs randomly select UGVs as access points (Random) and another scheme is the number of UAVs selected different UGVs is equal (Average).

B. Numerical Evaluation

Fig. 2 presents the convergence of the evolutionary game in the UAVs group. We set the initial group states in the UAVs group are $\{0.1, 0.2, 0.7\}$. With the dynamic process of the evolutionary game among UAVs, after several simulation steps, there are 50% UAVs selecting UGV 1 as the access point, 25% UAVs selecting UGV 2 as the access point and 25% UAVs selecting UGV 3 as the access point, respectively. During the evolutionary game process, each UAV compares its utility with the average utility of the group at each iteration step. The UAV would change its access selection if it finds its utility is smaller than the average utility. Then, it chooses the UGV as a new access point whose connected UAV's utility in the same group is larger than its own. In the end, all the UAVs in the same group have equal utility and the evolutionary game obtains the equilibrium solution.

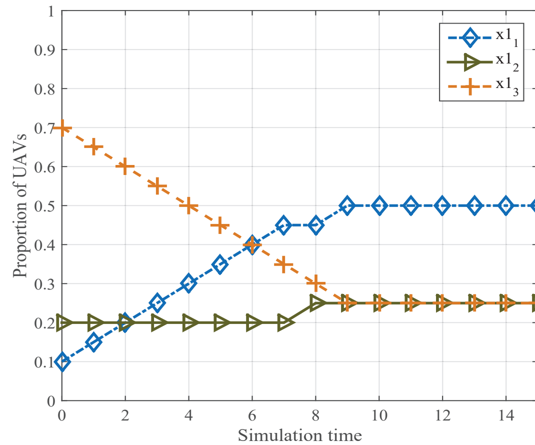

Fig. 2. The convergence of the evolutionary game of UAVs' group.

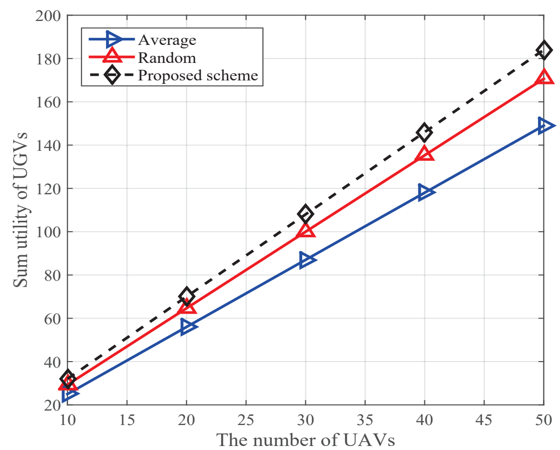

Fig. 3. Sum utility of UGVs vs. The number of UAVs.

Fig. 3 depicts the comparison of the proposed scheme with other schemes on UGVs' sum utility. In Fig. 3, the proposed scheme for the UGVs has a better performance than the other two schemes. Because the UGVs in our scheme compete with other players to obtain the optimal strategies so that they can maximize their own utility adaptively. This proves that the non-cooperative game among UGVs can help them achieve their optimal solutions. In addition, Fig. 3 indicates that the performance difference between our proposed scheme and the other two schemes gradually increases with the rise of the number of UAVs. Since the number of UAVs in the disaster area increases, the competition among UGVs is much fiercer

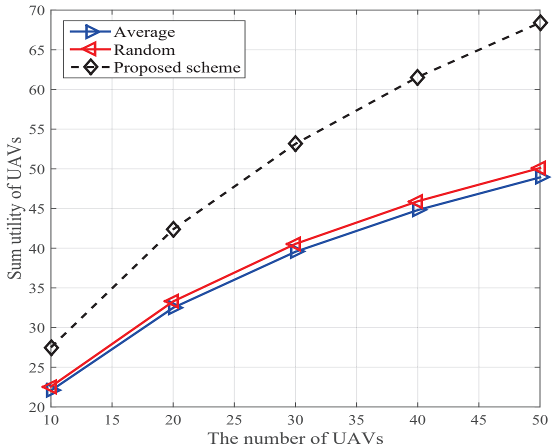

Fig. 4. Sum utility of UAVs vs. The number of UAVs.

in order to obtain much larger utilities.

Fig. 4 indicates that the comparison of our proposed scheme with other schemes on UAVs' sum utility. Specifically, the sum utility of UAVs in the proposed scheme is always larger than the other two schemes. Moreover, the performance difference between our proposed scheme and the other two schemes substantially increases with the rise of UAVs' number in the same group. In our proposed scheme, the UAVs will change selections of access points if individual utility is smaller than the average utility of the entire group. In the end, all the UAVs in the same group have equal utility in the same group. Therefore, the evolutionary game we adopted to formulate the dynamic process of UAVs selecting access points is efficient.

VII. Conclusion

In this paper, we have investigated the problem of bandwidth allocation in the emergency communication network for disaster relief. Since the original communication infrastructures are damaged by disasters, we have introduced multi-UAVs and multi-UGVs to build a disaster relief network architecture. Then, we use the Stackelberg game to solve the challenge of transmitting rescuing data with limited bandwidth. In detail, the Stackelberg game is exploited to model the interactions between UGVs and UAVs, where UGVs are leaders to decide the bandwidth amount and UAVs are followers to determine the strategies for access selection. Besides, the access selection of UAVs has been expressed as an evolutionary game in order to maximize the transmission rate. And we have formulated the competition among UGVs by a non-cooperative game to obtain their optimal bandwidth capacity. Moreover, a backward induction method-based iterative algorithm is developed to derive the Stackelberg equilibrium. In the end, simulation results have proved that our scheme outperforms conventional schemes. In the future, we will study how to ensure the security of emergency communication networks.

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